

Completed HCI Co-Design Canvas

Preferred implementation: nurse-facing ED triage decision-support dashboard

AI-Assisted Emergency Department Triage

Mercer General Hospital - Week 9 Final Submission

1. Problem

2. Ethics

3. Guidelines

4. MVP

5. Environment

6. Form

Design principle: support the nurse's judgement; never replace it.

HCI - ED Triage Desk

Problem Space Co-Design Canvas - Week 9 Interim

User group(s)

- Primary: ED triage nurses
- Secondary: charge nurse, ED physicians, clinical IT, quality/governance staff
- Indirect: patients whose care is prioritised using the nurse's final decision

Characteristics

- Clinically trained; works under time pressure and frequent interruptions
- May be fatigued during 12-hour shifts
- Needs rapid, glanceable information and a reliable manual fallback

Needs

- A recommendation that is visible in seconds
- Clear missing-data and uncertainty warnings
- A fast confirm/override workflow with accountability
- No replacement of professional judgement

User goals

Short term

- Reduce time spent assembling pre-triage data
- Make high-acuity patterns easier to notice
- Standardise documentation of confirm/override decisions

Long term

- Support a safe silent pilot
- Improve consistency without deskilling nurses
- Create auditable evidence for local validation and future refinement

AI SYSTEM task(s)

Short term

- Pull or accept patient data
- Validate inputs and identify missing values
- Display proposed ESI level 1-5 with text, icon, colour and confidence
- Record nurse confirmation or override reason

Long term

- Summarise aggregate use and override patterns
- Support review of performance, fairness, drift and workflow fit
- Never auto-finalise a clinical decision

Advantages / why this setting may benefit

Fast visual scanning

No physical movement near patients

Works with trained staff

Easy to log and review

Supports manual fallback

Preferred HCI Implementation - Ethical Considerations

ED triage desk, nurse-facing decision support

Physical safety

Problem

A delayed or confusing alert may contribute to delayed escalation.

Solution

Critical red flags remain visible; manual triage is always available; the interface never blocks care.

Data security

Problem

Patient details could be exposed through weak access control or screen visibility.

Solution

Role-based login, automatic timeout, encryption, audit logging and no patient data in browser logs.

Transparency

Problem

A numeric ESI output can appear more certain or authoritative than it is.

Solution

Show confidence status, missing inputs, data source and a clear 'nurse judgement is final' statement.

Equality across users

Problem

Model performance and interface access may vary by subgroup or disability.

Solution

No colour-only meaning; accessible interaction; subgroup monitoring and local fairness review before expansion.

Emotional consideration

Problem

An urgent alert can increase stress or imply that the nurse made an error.

Solution

Neutral, respectful wording; no blame language; support escalation without shaming the user.

Behaviour enforcement

Problem

Automation bias may pressure nurses to accept the model recommendation.

Solution

Confirm and override controls are equally visible; override requires a reason for audit, not punishment.

Preferred HCI Implementation - Design Guidelines

Guidelines that govern the ED triage desk interface

Environment guidelines

- Optimise for a noisy, bright, cluttered and interruption-heavy ED.
- Assume the user may be fatigued late in a 12-hour shift.
- Maintain a complete manual pathway during power or connectivity loss.
- Fit the existing desk without adding new physical obstruction.

Form guidelines

- Use high contrast, large type and large touch targets.
- Keep the proposed ESI level and missing-data state above the fold.
- Use text, icon and shape in addition to colour.
- One primary action per screen; avoid dense dashboards.

Interaction guidelines

- Show the recommendation only after required inputs are validated.
- Provide confirm and override from the same screen.
- Make manual mode reachable within two actions.
- Do not use routine audio; never hide system status.

Behaviour guidelines

- Use neutral, non-authoritative language.
- Never auto-finalise ESI or initiate treatment.
- When uncertain, ask for manual review rather than guessing.
- Log failures and overrides without penalising the nurse.

Preferred HCI Implementation - Minimum Viable Product

The smallest pilot that delivers value without replacing clinical judgement

Where and when

- Main ED triage desk
- Silent pilot during staffed shifts
- Nurse-facing only

System role

- Summarise valid inputs
- Propose ESI 1-5
- Flag uncertainty and missing data

Interaction tone

- Calm, concise and professional
- Supportive, never commanding
- Plain language

MVP features

- EHR pull with manual fallback
- Vital-sign and chief-complaint summary
- Proposed ESI with text + icon + colour
- Confidence / missing-data warning
- Confirm and override controls
- Audit log and final nurse decision

Connection to systems

- Read-only Mercer EHR interface
- Hospital authentication
- Audit-log storage
- Offline/manual mode
- No treatment orders or autonomous workflow actions

Pilot success measures

- Median response time below 1.5 seconds
- Manual fallback completed without confusion
- Nurses can identify recommendation and missing-data state within 3 seconds
- Override events are completely logged
- No adverse event attributable to the interface during the silent pilot

Preferred HCI Implementation - Environment Canvas

Mercer operating conditions and the design response

Noise

Operating condition

Ambient noise may exceed 70 dB; audio competes with existing alarms.

Design response

Use visual-first alerts; reserve audio for approved exceptions only.

Lighting

Operating condition

Fluorescent light, shadows and changing ambient conditions affect legibility.

Design response

High contrast, large type, matte display and no colour-only meaning.

Shift fatigue

Operating condition

Nurses may work 12-hour shifts and make decisions under interruption.

Design response

Short labels, one primary action, predictable screen order and visible system state.

Power instability

Operating condition

Generator switching and brief outages are foreseeable.

Design response

UPS-compatible workstation, safe recovery, offline banner and manual protocol.

Physical space

Operating condition

The desk is small, shared and cluttered.

Design response

Use existing screen; no extra hardware in the nurse's movement path.

Privacy

Operating condition

Patient details may be visible to people near the desk.

Design response

Role-based access, automatic timeout, privacy-aware layout and minimal on-screen identifiers.

Preferred HCI Implementation - Form Canvas

Outwardly perceptible qualities of the nurse-facing interface

ED TRIAGE DECISION SUPPORT

Patient: Encounter 00542

Input status: COMPLETE

Clinical inputs

HR 112 bpm
BP 98/62
RR 24/min
SpO2 91%
Temp 37.2 C

PROPOSED ESI

2
High acuity

Review required

Low oxygen saturation; confirm patient condition.

CONFIRM

OVERRIDE

AI recommendation is decision support. Nurse judgement is final.

Form decisions

Screen form

Existing nurse workstation; responsive web interface.

Visual hierarchy

Proposed ESI and missing-data state are the most prominent elements.

Colour use

Red/orange/yellow are paired with text, icon and shape.

Controls

Large confirm and override buttons; override reason appears only when needed.

Language

Short, neutral labels and plain international English.

Accessibility

High contrast, scalable text, keyboard navigation and no colour-only meaning.

Information density

Only action-critical data on the decision screen; full record available separately.